

Universal gates from braiding and fusing anyons on quantum hardware

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Chiu Fan Bowen Lo^{1✉}, Anasuya Lyons¹, Dan Gresh², Michael Mills², Peter E. Siegfried², Maxwell D. Urmey², Nathanan Tantivasadakarn³, Henrik Dreyer⁴, Ashvin Vishwanath¹, Ruben Verresen^{5✉} & Mohsin Iqbal^{4✉}

A quantum computer requires the ability to store and manipulate information globally to protect against local noise. Topologically ordered phases^{1,2} offer two routes: encoding information in the ground-state subspace³ or in anyonic excitations^{1,4,5}. The toric code¹ exemplifies the first approach but does not intrinsically support a universal gate set. The latter—topological quantum computation—implements gates by braiding non-Abelian anyons⁶ around each other. However, the simplest non-Abelian generalizations of the toric code cannot achieve universality by braiding alone^{7–9}. Here we demonstrate that anyon fusion, used as a computational primitive, renders these minimally non-Abelian topologically ordered states universal. We prepare a 54-qubit ground state of the quantum double of S_3 , the smallest non-Abelian group, on the H2 processor of Quantinuum. We encode logical information in the global fusion space of non-Abelian anyons, and by combining braiding with fusion, we realize a universal topological gate set and read-out, which we demonstrate by topologically preparing a magic state. This demonstrates that the S_3 topologically ordered state is scalably preparable, yet rich enough to support a universal gate set. More broadly, this work opens up new pathways for harnessing the intrinsic properties of quantum matter to manipulate quantum information.

The appeal of topologically ordered phases of matter lies in their ability to encode information in global properties insensitive to local perturbations^{1,2}. The toric code¹ exemplifies this idea: an emergent $\mathbb{Z}_2$ gauge theory¹⁰ whose anyonic quasiparticles exhibit non-trivial exchange statistics^{11,12}, forming the foundation of the paradigmatic surface code^{13,14}, where quantum information is stored nonlocally in the degenerate ground-state subspace³. A qualitatively different approach is known as topological quantum computation, in which information is directly encoded in the internal degrees of freedom of non-Abelian anyons^{1,4}, which is minimally realized in quantum double models based on finite non-Abelian groups¹.

Until recently, the implementation of topological quantum computation was out of experimental reach. However, the discovery that, for solvable groups, constant-depth adaptive circuits can prepare non-Abelian quantum double models^{15–19} as well as their anyons^{19–21} has prompted renewed theoretical interest in their utility^{22–26}. These measurement-based approaches substantially lower the experimental barriers to realizing these phases on quantum hardware, with a recent experimental realization of D_4 topological order²⁷.

Despite increased experimental accessibility to non-Abelian defects and topological orders^{27–33}, a fundamental limitation remains. Although topological orders based on finite groups are desirable because of being minimal non-Abelian generalizations of the toric code, their braiding operations do not provide a universal set of quantum gates^{7–9}. This

stands in contrast to more exotic topological phases such as Fibonacci topological order, in which braid-only universality can be achieved⁴, albeit at the cost of significantly greater complexity in their realization^{29,31}, in part because of conjectured no-gos for scalable preparation protocols¹⁵. Moreover, their irrational quantum dimensions complicate the search for leakage-free gates^{34–36}, in contrast to the integer quantum dimensions of anyons in quantum doubles. Moreover, although a finite decoding threshold has recently been proven for quantum doubles with solvable groups by leveraging the similarity with the toric code³⁷, no similar results exist for more exotic topological order.

Even in idealized settings for topological quantum computation, however, measurements of anyon type are required to extract information encoded in non-Abelian anyons^{1,38}. This observation raises a natural question: given measurement has already been proven instrumental for preparing and manipulating non-Abelian topological phases, can it also be harnessed to enhance their computational power? Theoretical work has suggested that topological orders associated with cyclic anyon fusion rules³⁹, of the form $a \times \bar{a} = a + \dots$, become universal when anyon fusion measurement is incorporated into the gate set^{8,9,40–44}. Although this cyclic criterion excludes the previously realized D_4 topological order, it applies to the quantum double of the smallest non-Abelian group, S_3 (refs. 41,45–47).

This work presents, to our knowledge, the first experimental realizations of a universal topological gate set based on the braiding and

¹Department of Physics, Harvard University, Cambridge, MA, USA. ²Quantinuum, Broomfield, CO, USA. ³C. N. Yang Institute for Theoretical Physics, Stony Brook University, Stony Brook, NY, USA.

⁴Quantinuum, Munich, Germany. ⁵Pritzker School of Molecular Engineering, University of Chicago, Chicago, IL, USA. ✉e-mail: chiufanbowenlo@g.harvard.edu; verresen@uchicago.edu; mohsin.iqbal@quantinuum.com

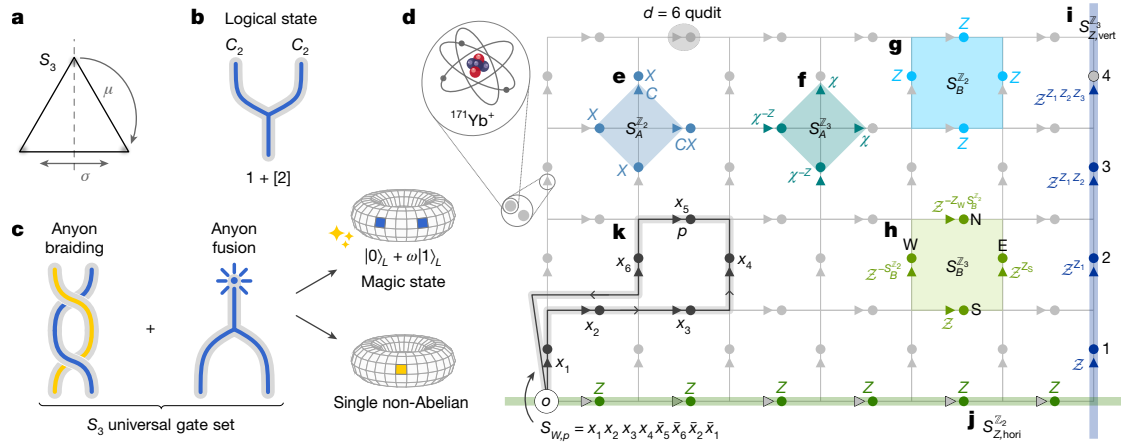


Fig. 1 | Topological order based on S_3 . **a**, Symmetry group $S_3 = \mathbb{Z}_3 \rtimes \mathbb{Z}_2$ of an equilateral triangle, generated by rotation μ and reflection σ . **b**, Logical state encoded in the flux-neutral fusion space of a pair of C_2 fluxes. **c**, Anyon braiding together with fusion realizes a universal gate set for quantum computation. **d**, The code is defined on a square lattice on a torus with oriented edges hosting $d = 6$ qudits, composed of a qubit (solid circle) and a qutrit (solid triangle). The triangle direction specifies the edge orientation. Physically, each qubit is encoded in the hyperfine states of $^{171}\text{Yb}^+$ ions. **e–k**, The stabilizers decompose into Pauli operators X, Z on qubits and generalized Pauli operators X, Z

together with charge conjugation C on qutrits (Supplementary Information section 1B). There are qubit vertex stabilizers $S_A^{Z_2}$ (e), qutrit vertex stabilizers $S_A^{Z_3}$ (these act on qutrits, conditioned on qubits) (f), qubit plaquette stabilizers ($S_B^{Z_2}$) (g) and qutrit plaquette stabilizers ($S_B^{Z_3}$) (h). There are also non-contractible versions of the $S_Z^{Z_3}$ (i) and $S_Z^{Z_2}$ stabilizers (j), which toggle between ground states. The stabilizer $S_{W,p}$ associated with the global flux W is defined as the product of S_3 group elements x_i along the directed loop in **k**, beginning at the origin o and encircling exactly one plaquette p (Supplementary Information section 2).

fusion of non-Abelian anyons in a scalable non-Abelian topological order, in particular the S_3 quantum double model on a trapped-ion quantum computing architecture⁴⁸. Using a two-step process, which can be interpreted as coupling a $\mathbb{Z}_2$ and $\mathbb{Z}_3$ toric code^{15–17,19}, this enables the realization of S_3 topological order ground states as well as the realization of a single non-Abelian anyon on a torus⁴⁹, which we link to the cyclicity of the anyon fusion rules. We nonlocally encode logical information in the internal states of non-Abelian qutrits, and demonstrate a universal gate set on these topological qutrits, consisting of three primitive operations: (1) an entangling gate implemented by coherent anyon braiding; (2) measurement of the topological qutrit in the X -basis; and (3) measurement of the topological qutrit in the Z -basis. Any quantum operation on pairs of qutrits (and therefore also qubits) can be synthesized from these three primitives^{41,47}. Although these primitives cannot be realized in a strictly transversal way^{50,51}, and we use linear-depth circuits with measurements for our demonstrations, they can be scalably implemented with a constant-depth adaptive circuit. We further demonstrate how to prepare reference states (the so-called ‘bureau of standards’⁵²) and how our primitives can be used to prepare a magic state on a topological qutrit (Fig. 1a–c).

The S_3 quantum double

The internal Hilbert space consists of six-state qudits labelled by the elements of the group $S_3 = \langle \mu, \sigma | \mu^3 = \sigma^2 = (\mu\sigma)^2 = 1 \rangle$ (Fig. 1a). We encode this as a qutrit–qubit pair $g = \mu^a \sigma^b$, where a is the qutrit and b the qubit value. On the Quantinuum hardware, the qutrit is itself composed of two qubits, with each qubit being encoded in an ion (Fig. 1d). Errors caused by leakage into the extra qubit state can be mitigated through heralding and post-selection (Supplementary Information section 4).

The S_3 quantum double⁴¹ ground states are stabilized by the following qubit and qutrit projectors:

$$A_v^{Z_2} = \frac{1}{2}(1 + S_A^{Z_2}), \quad A_v^{Z_3} = \frac{1}{3}(1 + (S_A^{Z_3} + \text{h.c.})), \quad (1)$$

$$B_p^{Z_2} = \frac{1}{2}(1 + S_B^{Z_2}), \quad B_p^{Z_3} = \frac{1}{3}(1 + (S_B^{Z_3} + \text{h.c.})), \quad (2)$$

where the qubit and qutrit vertex and plaquette stabilizers, $S_A^{Z_2}$, $S_A^{Z_3}$, $S_B^{Z_2}$, and $S_B^{Z_3}$ are defined in Fig. 1e–h. $S_A^{Z_2}$ ($S_A^{Z_3}$) represent the $\mathbb{Z}_2$ ($\mathbb{Z}_3$) X -type stabilizers imposing Gauss law constraints at a vertex, such that the corresponding projector $A_v^{Z_2}$ ($A_v^{Z_3}$) projects into the zero-charge sector at vertex v . The plaquette stabilizer $S_B^{Z_2}$ ($S_B^{Z_3}$) are Z -type stabilizers detecting $\mathbb{Z}_2$ ($\mathbb{Z}_3$) flux, such that the projector $B_p^{Z_2} B_p^{Z_3}$ projects into the trivial-flux sector at plaquette p . The resulting state can be interpreted as a deconfined S_3 gauge theory, whose quasiparticles exhibit anyonic exchange statistics due to the Aharonov–Bohm effect.

There are eight anyon types in the S_3 quantum double model (Supplementary Information section 1C). Here, we work with three types of non-Abelian anyons: the charge transforming under the two-dimensional $[2]$ representation of S_3 , detected by a violation of $A_v^{Z_3}$ (‘ $[2]$ charge’); the two-dimensional flux that violates a $B_p^{Z_3}$ plaquette, with flux state labelled by the conjugacy class C_3 represented by μ (‘ C_3 flux’); and the three-dimensional flux, which violates both $B_p^{Z_2, Z_3}$ plaquettes, with flux state labelled by the conjugacy class C_2 represented by σ (‘ C_2 flux’).

State preparation

Ground-state preparation

We use the all-to-all connectivity of the trapped-ion platform to realize the S_3 quantum double ground states on a torus, in which there are eight ground states in the S_3 quantum double, in one-to-one correspondence with the eight anyon types. We label the ground states by non-contractible line operators (Fig. 1i, j).

The state-preparation protocol mirrors the decomposition of S_3 group elements into a qutrit and a qubit, using the fact that S_3 can be expressed as a semidirect product $\mathbb{Z}_3 \rtimes \mathbb{Z}_2$. We use a gauging procedure to prepare a $\mathbb{Z}_3$ qutrit toric code³⁰ and subsequently apply another gauging operation (that in isolation would produce a $\mathbb{Z}_2$ qubit toric code) to obtain the ground states of S_3 topological order¹⁷. We implement this construction using a fully unitary circuit (Supplementary Information section 6). The measured projector expectation values are shown in Fig. 2a (see Methods, section ‘Ground-state preparation’ for details of the expectation value estimation). Because the target state $|g_s\rangle$ is fully specified by its projectors, we bound the per-qudit

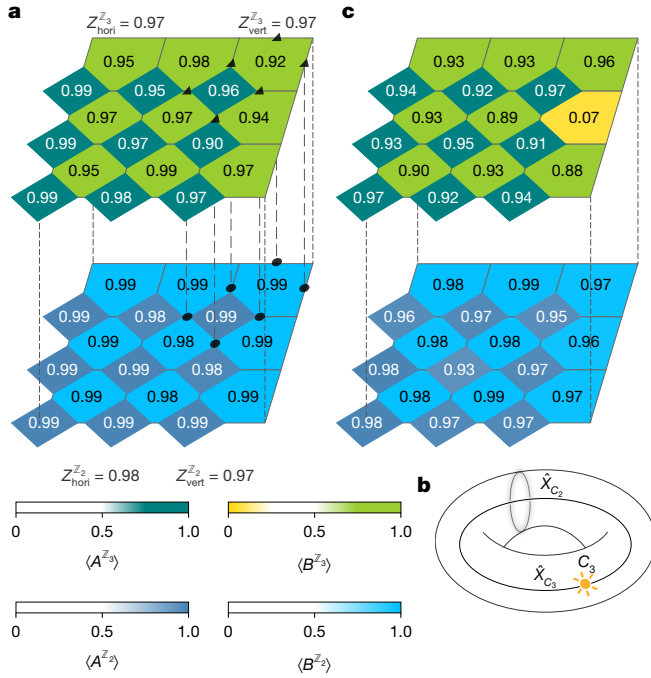


Fig. 2 | Ground and single-anyon states on 54 qubits. a, Z_2 (bottom) and Z_3 (top) vertex and plaquette projector expectation values on a 3×3 periodic square lattice (see equations (1) and (2)). Lighter squares denote plaquette projectors and darker rhombi denote vertex projectors. The mean Z_2 and Z_3 projector values are 0.987(2) and 0.962(4), respectively; the average standard error is 0.014, and the maximum standard error is 0.029. For the non-contractible projectors, the average (maximum) standard error is 0.017 (0.022). **b,** Single C_3 anyon creation by applying a horizontal $\hat{X}_{C_3}$ to a ground state with a vertical C_2 flux loop. **c,** The anyon is identified by a single excited B^{Z_3} plaquette projector. The average standard error is 0.020, and the maximum standard error is 0.034.

fidelity of the prepared state ρ on the 3×3 lattice with 18 sites as (Supplementary Information section 8)

$$0.970(5) \leq \langle \text{gs} | \rho | \text{gs} \rangle^{1/18} \leq 0.988(3). \quad (3)$$

As S_3 is a solvable group, it also admits a constant-depth adaptive preparation^{17,19}. We find a comparable fidelity between 0.930(8) and 0.978(2) (see Extended Data Fig. 2 and Supplementary Information section 6B1). We use the ground states prepared unitarily for the remaining demonstration because of the slightly better fidelity for the current system size; however, the measurement-based preparation protocol guarantees scalability of future experiments.

A single non-Abelian anyon on the torus

A hallmark of non-Abelian topological order is the existence of low-lying excited states hosting a single anyon on the torus. A recent realization of D_4 topological order demonstrated an excited state with a single unpaired Abelian anyon²⁷. Although such an unpaired anyon certifies the non-Abelianness of that state, here we target a more complex and computationally powerful quantum phase: a cyclic topological order. Cyclicity is expected to enable universal computation by braiding and fusion^{9,40}. A diagnostic of cyclic fusion rules is the ability to generate a single non-Abelian anyon on the torus by braiding an identical anyon pair around a non-contractible loop.

Braiding a C_2 flux pair around the vertical non-contractible cycle of the torus generates a non-trivial ground state with C_2 background flux. Starting from this non-trivial ground state, we can braid a C_3 flux pair around the horizontal non-contractible cycle (Fig. 2b). Owing to

the non-trivial braiding between the background C_2 flux and the C_3 anyons, the C_3 pair fuses to a single C_3 anyon after braiding around the horizontal cycle (Fig. 2c). This constitutes the experimental realization of a single non-Abelian flux anyon in a lattice model with topological order. See Supplementary Information section 7 for more details on the preparation protocol.

We note that we will not use this single non-Abelian to store quantum information. Rather, the above demonstrates the cyclicity of the fusion rules, which is a signature of the computational power of the S_3 quantum double. We next demonstrate explicit building blocks of a universal gate set.

Universal topological gate set

Logical qutrit encoding

To robustly store quantum information in nonlocal degrees of freedom, we encode it in the fusion space of spatially separated anyons. In particular, we use a pair of C_2 fluxes with trivial total flux (Fig. 1b), which ensures that the encoded information is unaffected by transporting pairs across the system. Despite its flux neutrality, the pair retains delocalized charge degrees of freedom due to the fusion rule $C_2 \times C_2 = 1 + [2] + \dots$. The associated fusion space already yields a two-dimensional logical subspace. Moreover, the internal structure of the $[2]$ charge sector is itself protected: the two charge states cannot be distinguished or toggled by any local operation that does not enclose both fluxes. We thus see that the separation distance between anyons gives us a code distance. Provided the fluxes remain well separated, we obtain a protected three-dimensional fusion space, realizing a topological qutrit.

We can label this logical qutrit in terms of the fluxes of the constituent C_2 fluxes:

$$|j\rangle_L = |\mu^j \sigma, \mu^j \sigma\rangle, \quad j = 0, 1, 2, \quad (4)$$

defining our Z -basis. Moreover, the nonlocal charges are diagonal in the logical X -basis:

$$|\tilde{j}\rangle_L = \frac{1}{\sqrt{3}}(|0\rangle_L + \omega^j |1\rangle_L + \bar{\omega}^j |2\rangle_L), \quad j = 0, 1, 2 \quad (5)$$

where $\omega = e^{2\pi i/3}$. It can be checked that the $|\tilde{0}\rangle_L$ flux pair is in the trivial anyon sector, whereas the $|\tilde{1}\rangle_L$ and $|\tilde{2}\rangle_L$ flux pairs behave as a (delocalized) $[2]$ charge.

For the flux labels in equation (4) to be well-defined for spatially separated C_2 anyons, we can consistently measure the global flux W associated with a loop based at a fixed-but-arbitrary choice of origin (see Fig. 1k for an example of a global flux). We call this the ‘absolute’ encoding. Although this is sufficient to define topological flux and demonstrate universal gate sets (see the following sections), the origin defines an Achilles’ heel through which quantum information can leak out by local charge measurement. This is avoided by a ‘relative’ encoding, where instead of using a fixed origin, the flux values are relative to other pairs. This requires building a ‘bureau of standards’⁵², which we demonstrate in the section ‘Bureau of standards’. We note that this approach is fully topological, in the sense that no individual local measurement can corrupt the logical information, and if so desired, it can be expressed in terms of fusion trees (Supplementary Information section 9).

In ref. 47, it is explained that three primitives (pull-through gate, Z -basis measurement and X -basis measurement) constitute a universal gate set. Here we demonstrate the experimental realization of these primitives. The braiding of non-Abelian fluxes is enabled by technical advancements in protocols to coherently move non-Abelian anyons using local flux label while preserving the global logical information (C. F. B. Lo et al., manuscript in preparation) (see Supplementary Information section 3 for theory overview). At the circuit level, the braiding

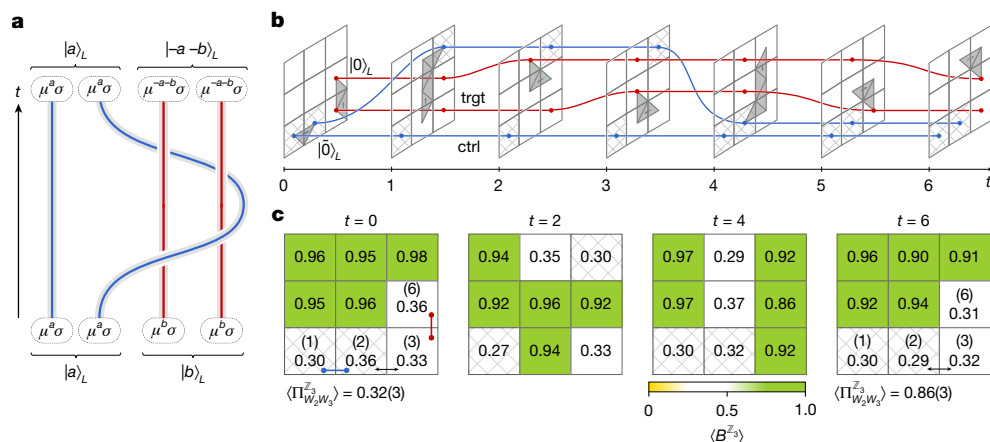


Fig. 3 | Pull-through gate entangles topological qutrits. a, Braiding between the logical pairs (equation (4)) gives an entangling gate (equation (6)) by the conjugation $(\mu^a \sigma) \mu^b \sigma (\mu^a \sigma)^{-1}$. **b**, Spacetime diagram of the protocol on the 3×3 torus. Two pairs of C_2 flux anyons, control (blue worldlines) and target (red worldlines), are initialized in the logical state $|\tilde{0}\rangle_L |0\rangle_L$. One endpoint of the control qutrit is braided (pulled through) around both endpoints of the target and returned to its initial position at $t = 6$. Oriented grey triangles indicate the ribbon operators implementing anyon motion. **c**, Expectation values of the

of non-Abelian anyons is implemented by linear-depth (in the distance the anyons travel) unitary circuits with one ancilla; explicit circuits can be found in Supplementary Information section 10.

Pull-through gate

We begin with the entangling primitive, implemented by braiding a constituent C_2 flux of one logical qutrit around the flux pair of another logical qutrit (Fig. 3a). This operation, known as the pull-through gate, exploits the non-Abelian braiding relations of the S_3 quantum double. In general, when fluxes braid around each other, they are each conjugated by their total flux. In this particular case, this leads to the logical two-qutrit gate

$$U|a, b\rangle_L = |a, -a-b\rangle_L. \quad (6)$$

We will demonstrate the action of this pull-through gate on $|\tilde{0}\rangle_L |0\rangle_L$, which creates a qutrit Bell state,

$$U|\tilde{0}\rangle_L |0\rangle_L = \frac{1}{\sqrt{3}}(|0, 0\rangle_L + |1, 2\rangle_L + |2, 1\rangle_L), \quad (7)$$

where the logical information is entangled across the flux pairs (Fig. 3b). Figure 3c shows the evolution of the B^{Z_3} projectors expectation values during the protocol (see Methods, section ‘Pull-through gate’ for protocol details and Extended Data Fig. 3 for further data). The B^{Z_3} projectors identify the locations of logical qutrit endpoints, revealing local information but not the encoded logical state.

To verify the resulting entanglement between the two flux pairs, we certify correlations in the Z -basis, quantified by $\langle W_{p_1}^{Z_3} W_{p_2}^{Z_3} \rangle$, where W_{p_i} denotes the topological W -flux operator encircling plaquette p_i (see Supplementary Information section 2C for the precise path). The observed increase of this correlator from its initial value at $t = 0$ to near unity at $t = 6$, as both qutrits return to their original lattice positions, provides strong evidence that the prepared state closely approximates the ideal target state.

X -basis measurement

Apart from braiding logical anyons, fusion of anyons also implements logical operations, in addition to providing a read-out mechanism. The partial X -basis measurement, which distinguishes between the charge-free $|\tilde{0}\rangle_L$ state and the charged $\{|\tilde{1}\rangle_L, |\tilde{2}\rangle_L\}$ subspace, is implemented

local projectors B^{Z_3} remain close to 1 except on plaquettes hosting logical qutrit endpoints, where they are approximately $1/3$; the average (maximum) standard error is 0.028 (0.044). The nonlocal correlator $\langle \Pi_{W_2 W_3}^{Z_3} \rangle$ certifies Z correlations between the control and target qutrits. Its measured value is consistent with the predicted value of $1/3$ at $t = 0$ and increases to 0.86(3) at $t = 6$, in good agreement with the theoretical value of 1 for the prepared qutrit Bell state.

by braiding and fusion of C_2 fluxes (Fig. 4a). This coherent detection of $|\tilde{0}\rangle_L$ is sufficient for a universal gate set on the $\{|0\rangle_L, |1\rangle_L\}$ qubit⁴⁷.

We demonstrate the X -basis measurements on two input states: $|\tilde{0}\rangle_L$ and $|\tilde{1}\rangle_L$. The $|\tilde{0}\rangle_L$ state is charge neutral, whereas the $|\tilde{1}\rangle_L$ state stores a $[2]$ charge nonlocally (that is, the flux pair fuses to a $[2]$ charge). Therefore, the way to measure in the X -basis is by braiding with an ancilla C_2 flux pair and checking the fusion outcome: if the pair fuses to vacuum, then we have measured in the $|\tilde{0}\rangle_L$ state, whereas if the pair fuses to a $[2]$ charge, then it is orthogonal to the $|\tilde{0}\rangle_L$ state. In the latter case, we can repeat the braiding to remove the $[2]$ charge.

The projection violations for the input states $|\tilde{0}\rangle_L$ are shown in Fig. 4b (top row) and $|\tilde{1}\rangle_L$ (bottom row) (see Extended Data Fig. 4 for further data and Supplementary Information section 10B for implementation details). For the $|\tilde{0}\rangle_L$ input state, the vertex projector A^{Z_3} is not violated after fusion (maintaining a value close to 1), demonstrating that we have measured the state to be in the $|\tilde{0}\rangle_L$ state. For the $|\tilde{1}\rangle_L$ input state, A^{Z_3} is violated after the fusion (with the stabilizer value close to 0), indicating the presence of a $[2]$ charge at the vertex. This demonstrates that the state is in the subspace spanned by the $|\tilde{1}\rangle_L, |\tilde{2}\rangle_L$ state.

Z -basis measurement

As the final primitive that completes the universal gate set, the Z -basis measurement is a comparison measurement: the measurement outcome indicates whether the data state is the same as the reference state or orthogonal to it. At the physical level, the Z -basis measurement determines the flux content of the logical flux pair by braiding and fusion of a $[2]$ charge pair around one flux each from the data and the reference pair (Fig. 4c). If the data and reference flux pairs are in the same internal state, then the $[2]$ charge pair winds around a trivial flux and fuses back to vacuum. If the data and reference flux pairs are in different internal states, then the total flux is a C_3 group element, which results in non-trivial braiding: there is a $3/4$ probability of obtaining a remnant $[-]$ charge after fusion.

We first instantiate the data logical state (here $|0\rangle_L$) in the middle row and the reference logical states $|0\rangle_L$ and $|1\rangle_L$ in (Fig. 4c, bottom and top rows). We then braid $[2]$ charge pairs to do comparison measurement of the data flux pair with the reference pairs; the projection violations are shown in Fig. 4d (see Extended Data Fig. 5 for further data and Supplementary Information section 10C for implementation details). Comparing the data state with the $|1\rangle_L$ state, there is a violation

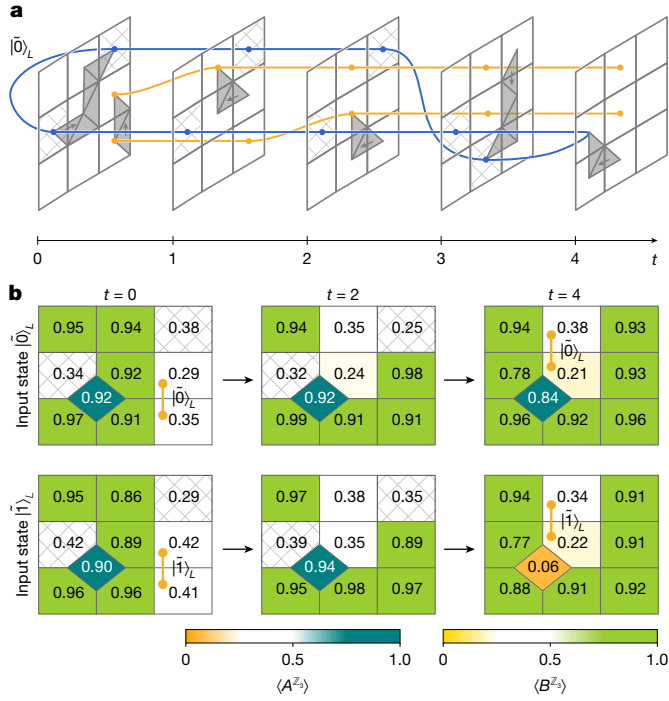


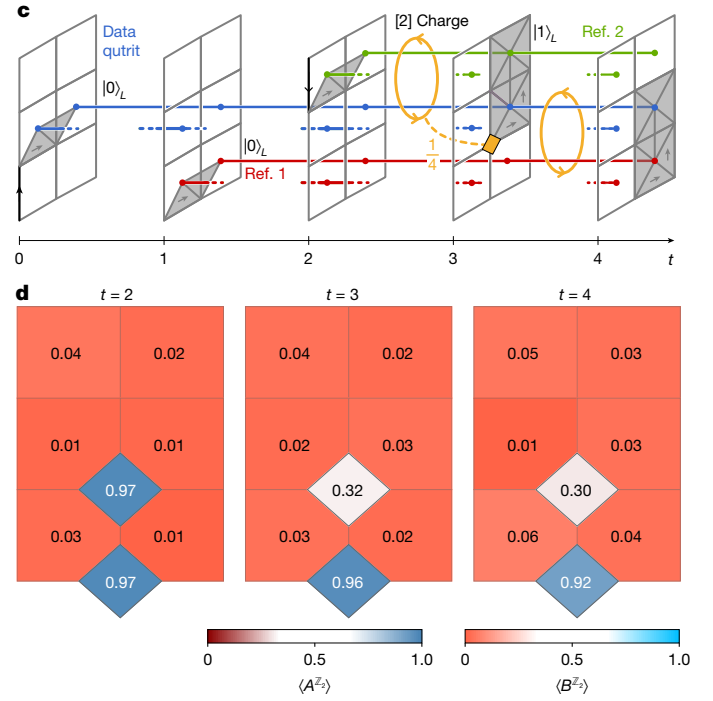
Fig. 4 | Logical $\mathcal{X}$ -basis and $\mathcal{Z}$ -basis measurements. **a**, For logical $\mathcal{X}$ -basis measurement, a symmetric ‘measurement’ flux pair (blue worldlines), created from the vacuum in the state $|\tilde{0}\rangle_L$, is braided around an endpoint of the flux pair of the input qutrit (yellow worldlines) and then fused. The fusion outcome diagnoses the input qutrit state: annihilation to the vacuum corresponds to $|\tilde{0}\rangle_L$, whereas a remnant charge anyon indicates the $\{|\tilde{1}\rangle_L, |\tilde{2}\rangle_L\}$ subspace. **b**, Experimental verification of the logical $\mathcal{X}$ -basis measurement protocol at $t = 0, 2, 4$. Top row indicates input state $|\tilde{0}\rangle_L$, and bottom row indicates $|\tilde{1}\rangle_L$. Local stabilizers remain insensitive to the logical content of the qutrits, with B^{Z2} values near $1/3$ at all flux endpoints (average/maximum standard error $0.034/0.059$). At fusion ($t = 4$), the vertex projector A^{Z2} yields $0.84(4)$ for $|\tilde{0}\rangle_L$, consistent with vacuum annihilation, and $0.06(3)$ for $|\tilde{1}\rangle_L$, indicating a remnant charge anyon. **c**, Schematic of the logical $\mathcal{Z}$ -basis measurement protocol. **d**, Experimental demonstration of the logical $\mathcal{Z}$ -basis measurement protocol. Local plaquette projectors B^{Z2} remain near zero at $t = 2, 3, 4$, confirming persistent qutrit endpoints (average/maximum error $0.014/0.021$). At $t = 3$, the vertex projector A^{Z2} at the first fusion site yields $0.32(4)$, consistent with the expected value $1/4$; at $t = 4$, A^{Z2} at the second fusion site is $0.92(2)$, indicating deterministic annihilation to the vacuum.

in the vertex projector A^{Z2} , where the $[2]$ charge pair is fused at $t = 3$ (Fig. 4d), indicating that the data qutrit is not in the $|\tilde{1}\rangle_L$ state. Finally, comparing the data state and the $|\tilde{0}\rangle_L$ state, there is no additional violation in the A^{Z2} vertex projector (at $t = 4$ in Fig. 4d), supporting that the data flux pair is in the $|\tilde{0}\rangle_L$ state. This comparison measurement between logical states also allows us to transition from the absolute encoding (equation (4)) to the relative encoding.

Bureau of standards

To define the relative encoding, we require a bureau of standards, as mentioned in section ‘Logical qutrit encoding’. Here we demonstrate the preparation of states in the $\mathcal{Z}$ -basis bureau of standards (the procedure is shown schematically in Extended Data Fig. 6a). Three charge-neutral $|\tilde{0}\rangle_L$ states are instantiated by pair-creation from the vacuum. Without loss of generality, we designate the lowest C_2 flux pair as the reference logical $|\tilde{0}\rangle_L$ state, and define all other logical states relative to this reference. Then, $\mathcal{Z}$ -basis comparison measurements are performed among the three possible pairings. By post-selecting on the $[-]$ fusion channel using mid-circuit measurements, indicating each of the three pairs occupies distinct $\mathcal{Z}$ -basis states, the reference $|\tilde{1}\rangle_L$ and $|\tilde{2}\rangle_L$ state are designated. This completes the preparation of the $\mathcal{Z}$ -basis bureau of standards, certified by the inter- and intra-pair correlations (Extended Data Fig. 6b–d).

In the experiment, for simplicity, we have post-selected for the $[-]$ charge outcome for all three $\mathcal{Z}$ -basis measurements to obtain all three logical states. However, post-selection is not necessary. If the



Worldlines on the 3×2 lattice for a data qutrit (blue) and two reference qutrits: reference 1 (red) in $|\tilde{0}\rangle_L$ and reference 2 (green) in $|\tilde{1}\rangle_L$. A $[2]$ charge pair (yellow) encircles the data and reference-2 qutrits between $t = 2$ and $t = 3$; because the logical states differ, the pair annihilates to the vacuum with probability $1/4$ or leaves a remnant charge with probability $3/4$. Between $t = 3$ and $t = 4$, a second $[2]$ charge pair encircles the data and reference-1 qutrits, both in $|\tilde{0}\rangle_L$, resulting in deterministic vacuum annihilation. **d**, Experimental demonstration of the logical $\mathcal{Z}$ -basis measurement protocol. Local plaquette projectors B^{Z2} remain near zero at $t = 2, 3, 4$, confirming persistent qutrit endpoints (average/maximum error $0.014/0.021$). At $t = 3$, the vertex projector A^{Z2} at the first fusion site yields $0.32(4)$, consistent with the expected value $1/4$; at $t = 4$, A^{Z2} at the second fusion site is $0.92(2)$, indicating deterministic annihilation to the vacuum.

measurement outcome is not the $[-]$ fusion channel but the vacuum fusion channel, that implies we have partially prepared the bureau of standards (we documented the other possible outcomes in Extended Data Fig. 6c). We can repeat until success: create fresh C_2 flux pair in the $|\tilde{0}\rangle_L$ state and compare until we obtain the $[-]$ fusion outcome to prepare the full $\mathcal{Z}$ -basis reference states.

Demonstrating magic

To demonstrate the power of the S_3 universal gate set that we have constructed, we exhibit a simple procedure to create a magic state using topological operations. Starting with the basis state $|\tilde{1}\rangle_L \propto |\tilde{0}\rangle_L + \omega |\tilde{1}\rangle_L + \bar{\omega} |\tilde{2}\rangle_L$, we use the $\mathcal{Z}$ -basis measurement primitive to find this in ‘not $|\tilde{2}\rangle_L$ ’, collapsing the state into $|\tilde{0}\rangle_L + \omega |\tilde{1}\rangle_L$ (see Fig. 5 for the protocol). This gives a minimal demonstration of how measurement on qutrits enables the creation of non-Clifford states using only topological operations.

To certify the magic state, we measure the state in both $\mathcal{Z}$ - and $\mathcal{X}$ -basis. First, in the $\mathcal{Z}$ -basis, the resulting state has equal weights only on the $|\tilde{0}\rangle_L, |\tilde{1}\rangle_L$ basis states as expected (Fig. 5c). For the $\mathcal{X}$ -basis, we use the Achilles’ heel of the absolute encoding to our advantage: by interrogating the logical charge violations at the origin, we can infer the $\mathcal{X}$ -basis measurement outcome (see Table 1 for theory prediction and Methods, section ‘Charge measurement at origin’); in particular, the charge measurement at the origin is sensitive to the relative phase ω . In Fig. 5c (inset), the experimental value of the charge violation weight

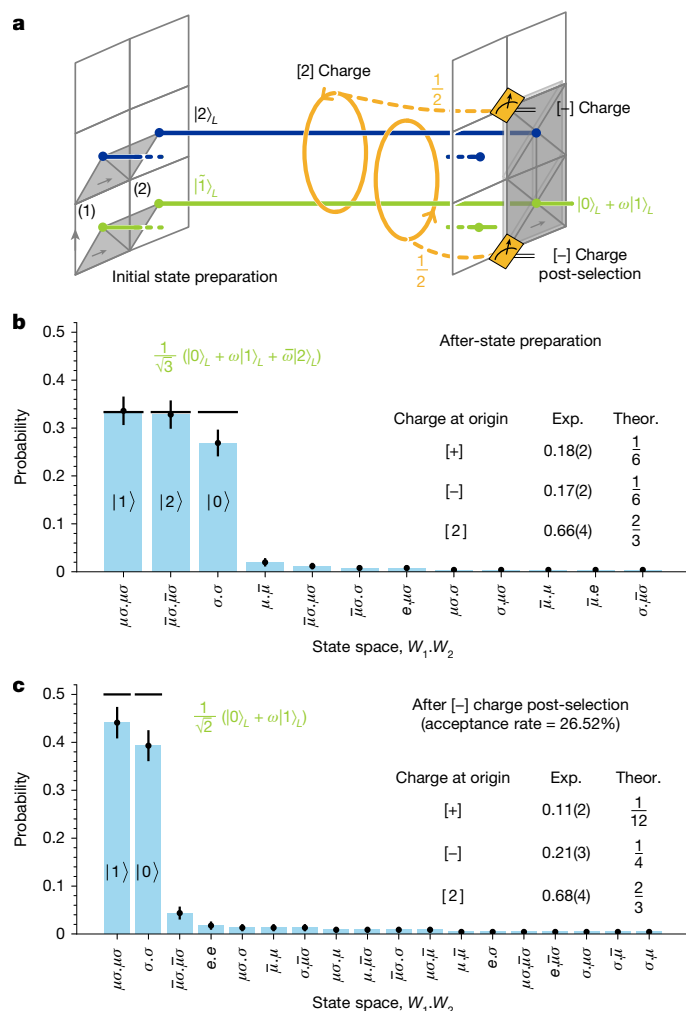


Fig. 5 | Illustrating universality by creating a topological magic state.

a, Spacetime diagram illustrating the procedure for preparing the magic state $\frac{1}{\sqrt{2}}(|0\rangle_L + \omega|1\rangle_L)$ (see Methods, section ‘Magic-state creation protocol details’ for preparation details). **b, c**, The bar plots show the joint probability of measuring the W -fluxes around plaquettes (1) and (2); **b** for the intermediate state $\frac{1}{\sqrt{2}}(|0\rangle_L + \omega|1\rangle_L + \omega|2\rangle_L)$ and **c** for the final state $\frac{1}{\sqrt{2}}(|0\rangle_L + \omega|1\rangle_L)$. Following initial state preparation, the three possible flux configurations, $\sigma, \sigma, \mu\sigma$ and $\bar{\mu}\sigma, \bar{\mu}\sigma$, which correspond to the logical basis states, occur with approximately equal probability of $1/3$. Successful preparation of the magic state is validated in the $\mathcal{Z}$ -basis by a marked increase in the probability of the σ, σ and $\mu\sigma, \mu\sigma$ flux configurations, accompanied by a suppression of the $\bar{\mu}\sigma, \bar{\mu}\sigma$ configuration; inset tables show that charge measurements at the origin in the experiments match with theoretical predictions, validating the state in the $\mathcal{X}$ -basis.

at the origin matches the theory prediction for when the state is $|0\rangle_L + \omega|1\rangle_L$, thus confirming the coherence of the magic state created.

Conclusion

In this work, we demonstrate a purely topological approach to universal quantum computation on quantum hardware. Rather than encoding information in a ground-state subspace, as in conventional topological codes, we encode logical states directly in the nonlocal fusion space of non-Abelian anyons, and we use their braiding and fusion as computational primitives. These abilities are enabled by the realization of a deconfined S_3 gauge theory, the non-Abelian topological order based on the smallest non-Abelian group. Exploiting its cyclic anyon fusion rules, we trap a single non-Abelian anyon on the torus, implement a universal logical gate set and demonstrate the

Table 1 | Theory prediction for the weights of charge violations at the origin for the magic state $|0\rangle_L + \omega|1\rangle_L$ and other possible states

State	1	[-]	[2]
$ \psi_\omega\rangle = \frac{1}{\sqrt{2}}(0\rangle_L + \omega 1\rangle_L)$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{2}{3}$
$ \psi_+\rangle = \frac{1}{\sqrt{2}}(0\rangle_L + 1\rangle_L)$	$\frac{1}{3}$	0	$\frac{2}{3}$
$\rho = p 0\rangle\langle 0 _L + (1-p) 1\rangle\langle 1 _L$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{2}{3}$

The state without a relative phase ($|\psi_+\rangle$) and the completely decohered state (ρ) have distinct weights of charge violations.

topological preparation of a magic state. These experimental demonstrations are also enabled by theoretical advances: (1) we showed how a two-anyon encoding is sufficient in contrast to proposed four-anyon encodings^{46,47}; (2) we developed coherent moving protocols to move non-Abelian fluxes deterministically; and (3) we showed that the logical encoding based on bureau of standards can be described in the fusion-tree perspective. Together, these results establish non-Abelian anyons as programmable and computationally powerful degrees of freedom, while the solvability of S_3 supports scalable preparation and control.

The realization of these abilities reflects a convergence of advances in quantum hardware and theoretical understanding of non-Abelian topological phases. Improvements in qubit number and coherence now enable the preparation and manipulation of complex topological states, whereas recent theoretical progress in lattice-scale anyon creation^{19–21}, encoding⁴⁷ and coherent moving (C.F.B.L. et al., manuscript in preparation) establish a setting in which non-Abelian anyons can be controlled rather than merely detected.

Looking ahead, encoding logical information directly in non-Abelian anyonic excitations opens new directions for fault-tolerant quantum computation. Owing to their close similarity to the surface code, non-Abelian quantum double models offer a promising route to extending surface code techniques while enabling new abilities, including universal topological gate sets. As our gate set is topological, the operations are protected as long as one is in the phase. Stabilizing the topological phase requires active error correction, which is beyond the scope of the present work, although a recent work proved a finite decoding threshold even with imperfect measurement during braiding³⁷. Numerical works indicate thresholds may be comparable to those of the surface code^{53,54}, and a systematic comparison between ground-state and fusion-space encoding is an exciting open direction. Our proof-of-principle results lead the way for more detailed resource estimates for the non-Abelian architecture, which will be crucial to evaluate how non-Abelian computing compares with more conventional stabilizer code schemes.

More generally, our work highlights how finite-group non-Abelian topological orders offer a compelling balance between computational power and experimental accessibility. Although more exotic phases may admit braid-only universality, solvable gauge theories such as S_3 occupy a ‘sweet spot’: they are sufficiently rich to support universal computation, yet sufficiently structured to admit efficient preparation and coherent anyon transport. S_3 quantum double provides a versatile platform for studying both scalable quantum computation beyond the stabilizer paradigm and new regimes of real-time dynamics of lattice gauge theories.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions

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Ground-state preparation

We prepare the ground state with trivial flux along both non-contractible loops of the torus. At the end of the preparation circuit, we non-destructively measure an implicitly prepared $S_A^{\mathbb{Z}_2}$ stabilizer together with an adjacent $S_B^{\mathbb{Z}_2}$ stabilizer. Empirically, we find that, in the event of a state-preparation error, these stabilizers are more likely to be violated³⁰. We, therefore, discard any experimental shot in which either of the two non-destructive measurement outcomes is -1 . Furthermore, we exploit the qutrit encoding—implemented using two physical qubits per qutrit—to detect errors that take qutrits outside the computational subspace; this serves as an independent heralding condition (Supplementary Information section 4).

Before destructive read-out of the edge qudits, a circuit barrier is inserted to ensure complete creation of the ground state. Expectation values of stabilizer and logical projectors are obtained from destructive measurements of each qutrit–qubit pair in the following bases:

- Z and $\mathcal{Z}$, yielding $B^{\mathbb{Z}_2}$, $B^{\mathbb{Z}_3}$ and horizontal and vertical logical Z projectors;
- Z and $\mathcal{X}$, yielding $A^{\mathbb{Z}_3}$ and $B^{\mathbb{Z}_2}$ projectors; and
- X and $\mathcal{C}$, yielding $A^{\mathbb{Z}_2}$ projectors.

Mean values of logical Z projectors of the S_3 quantum double are obtained by averaging across rows for the horizontal logical operators $Z_{\text{hor}}^{\mathbb{Z}_2}$ and $Z_{\text{hor}}^{\mathbb{Z}_3}$, and across columns for the vertical logical operators $Z_{\text{vert}}^{\mathbb{Z}_2}$ and $Z_{\text{vert}}^{\mathbb{Z}_3}$. After applying all heralding criteria, approximately 24% of shots are discarded. For the smaller 3×2 lattice, shown in Extended Data Fig. 1, the corresponding discard rate is about 13%.

For comparison, we also prepare the ground state using an implementation with mid-circuit measurements and feedforward (Supplementary Information section 6B1), which yields slightly lower fidelities (Extended Data Fig. 2).

Pull-through gate

First, we show that the action of flux braiding (conjugation by the total flux) reproduces the qutrit action of the pull-through gate as specified in the main text. We start with the state $|\tilde{0}\rangle_L|0\rangle_L = (|\sigma, \sigma\rangle + |\mu\sigma, \mu\sigma\rangle + |\bar{\mu}\sigma, \bar{\mu}\sigma\rangle)|\sigma, \sigma\rangle$. At the level of the flux, the fluxes are conjugated by the total flux participating in the braiding:

$$\begin{aligned} & U(|\sigma, \sigma\rangle + |\mu\sigma, \mu\sigma\rangle + |\bar{\mu}\sigma, \bar{\mu}\sigma\rangle)|\sigma, \sigma\rangle \\ &= |\sigma, \sigma\rangle|\sigma, \sigma\rangle + |\mu\sigma, \mu\sigma\rangle|\bar{\mu}\sigma, \bar{\mu}\sigma\rangle + |\bar{\mu}\sigma, \bar{\mu}\sigma\rangle|\mu\sigma, \mu\sigma\rangle \\ &= |0\rangle_L|0\rangle_L + |1\rangle_L|2\rangle_L + |2\rangle_L|1\rangle_L. \end{aligned} \quad (8)$$

In implementing the pull-through gate, as well as all other experiments presented in this work, logical qutrit states are prepared in the $\mathcal{X}$ -basis (that is, $|\star\rangle_L$) using an ancilla qutrit (Supplementary Information section 10A). Given the constrained budget of available qutrits (56 in total), ancilla qutrits are obtained by using a qubit-reuse technique, in which physical qubits that are no longer acted on in subsequent stages of the circuit are measured early and reset for use as ancillas.

After successful preparation, the ancilla qutrit is disentangled and ideally returns to the qutrit zero state. Measurement outcomes corresponding to a nonzero ancilla state—predominantly arising from memory errors in deeper circuits (Extended Data Table 1)—are therefore used to herald unsuccessful logical qutrit preparation, and the corresponding shots are discarded.

The correlator $\langle W_{p_1}^{\mathbb{Z}_3} W_{p_2}^{\mathbb{Z}_3} \rangle$, which we use to quantify correlations, ensures that

- when the control global flux is σ , the target is σ ;
- when the control global flux is $\mu\sigma$, the target is $\bar{\mu}\sigma$; and
- when the control global flux is $\bar{\mu}\sigma$, the target is $\mu\sigma$.

See Extended Data Fig. 3 for further data on non-trivial $\mathcal{Z}$ correlators.

Magic-state creation

Magic-state creation protocol details. The protocol begins by initializing the lower flux pair to be in the input state $|\hat{1}\rangle_L = \frac{1}{\sqrt{3}}(|0\rangle_L + \omega|1\rangle_L + \bar{\omega}|2\rangle_L)$ and the middle flux pair with the reference state $|2\rangle_L$. Two $\mathcal{Z}$ -basis comparison measurements with respect to $|2\rangle_L$ are performed using $[2]$ charges. After braiding the first $[2]$ charge pair around the end of both flux pairs, we post-select on the $[-]$ fusion outcome. This implies that the measured state is orthogonal to the $|2\rangle_L$ state, so that the state is projected to the orthogonal subspace. To ensure the correct coherence of the state in the subspace spanned by $|0\rangle_L$ and $|1\rangle_L$, we need to ensure no net charge is extracted from the flux pair that constitutes the logical state. With only one round of $[2]$ charge braiding, there is one local $[-]$ charge in the system, and by neutrality, the flux pair is storing another $[-]$ charge nonlocally. Therefore, we repeat the $\mathcal{Z}$ comparison measurement and wind another $[2]$ charge pair and again post-select on the $[-]$ fusion outcome, such that there are two local $[-]$ charges in the system; this ensures that the charge content of the resulting state is correct such that we obtain the state $|0\rangle_L + \omega|1\rangle_L$ (Fig. 5, bottom). See Extended Data Fig. 7 for further data and Supplementary Information section 10E for circuit-level implementation details.

Charge measurement at origin. Recall from section ‘ $\mathcal{X}$ -basis measurement’ that the $\mathcal{X}$ -basis measurement determines whether the state is $|\tilde{0}\rangle_L$ or in the orthogonal subspace by measuring the $[2]$ charge of the flux pair. In the absolute logical state, the origin contains the charge information of the logical state. This is due to the neutrality condition, the charges at the origin are exactly the partners of the charges shared by the flux pairs, such that fusing all anyons will return to vacuum. Therefore, by interrogating the logical charge violations at the origin, we can infer the $\mathcal{X}$ -basis measurement outcome.

The prediction for the expected charge violation weights for the $|0\rangle_L + \omega|1\rangle_L$ magic state and other possible states is shown in Table 1.

Data availability

The data generated in this study are available at Zenodo⁵⁵ (<https://doi.org/10.5281/zenodo.18054264>) under open access.

Code availability

The code used for numerical simulations is available at Zenodo⁵⁵ (<https://doi.org/10.5281/zenodo.18054264>).

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Author contributions C.F.B.L. and M.I. wrote the code generating the circuits for all experiments. The experiment was built and carried out by D.G., M.M., P.E.S. and M.D.U. The data analysis and interpretation were done by C.F.B.L., M.I., A.L., N.T., H.D., A.V. and R.V. C.F.B.L., A.L., N.T., A.V. and R.V. contributed to the ideation, theory and experiment design, including the definition of the operators for creation, coherent moving, and fusion of anyons; in particular, C.F.B.L., A.L. and R.V. contributed to the theory for the bureau of standards for the relative logical encoding. C.F.B.L. drafted the initial paper, which was refined by contributions from all authors, especially M.I., A.L., N.T., A.V. and R.V.; C.F.B.L., M.I. and A.L. contributed to the Supplementary Information; in particular, C.F.B.L. contributed to the protocol implementations, M.I. contributed to state preparation protocols and circuit compilation techniques, and A.L. contributed to the bureau of standards theory.

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Competing interests H.D. is a shareholder of Quantinuum. All other authors declare no competing interests.

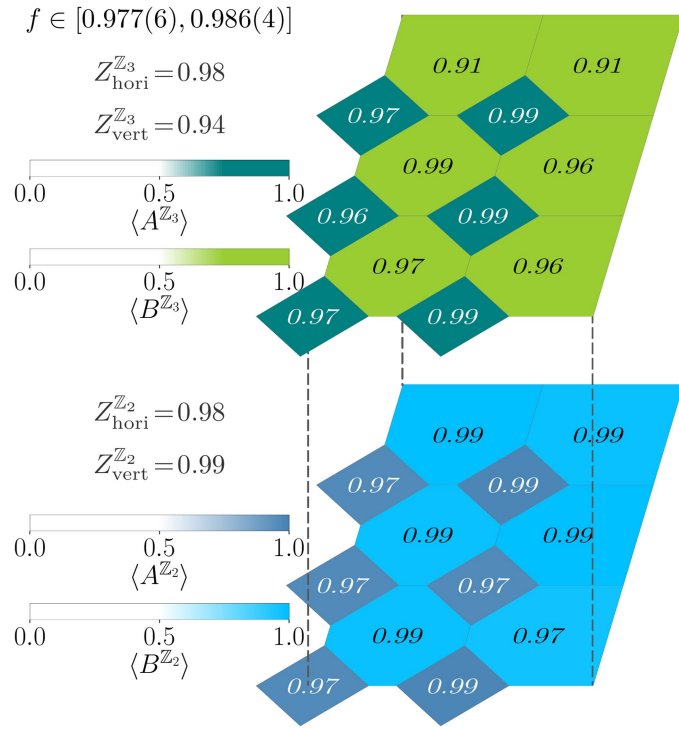
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-026-10709-y>.

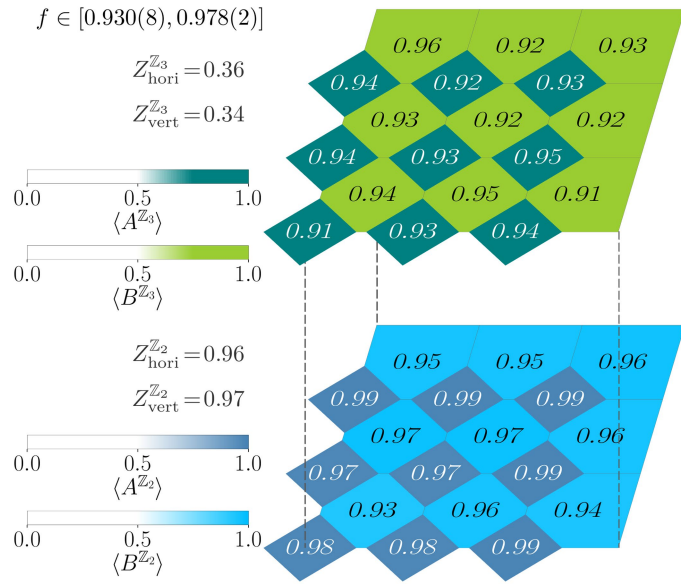
Correspondence and requests for materials should be addressed to Chiu Fan Bowen Lo, Ruben Verresen or Mohsin Iqbal.

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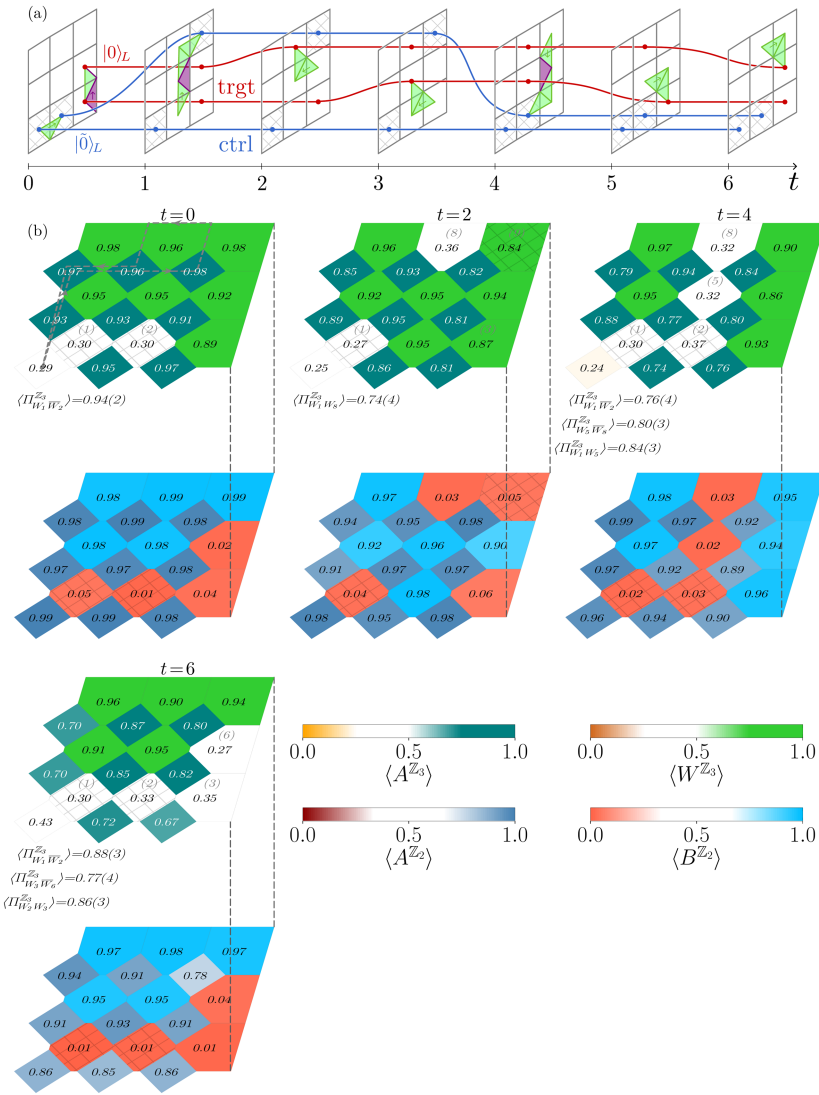
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Extended Data Fig. 1 | Ground state on the 3×2 lattice. Experimentally measured values of local and Z-logical projectors for a state prepared via a fully unitary protocol (Supplementary Information Section S6B2). The $\mathbb{Z}_3$ toric code is first prepared unitarily, followed by the unitary gauging of the $\mathbb{Z}_2$ charge-conjugation symmetry. The per-qudit fidelity of the resulting state is denoted by f . The average (maximum) standard error is 0.018 (0.034) for local projectors and 0.018 (0.031) for Z-logical projectors.

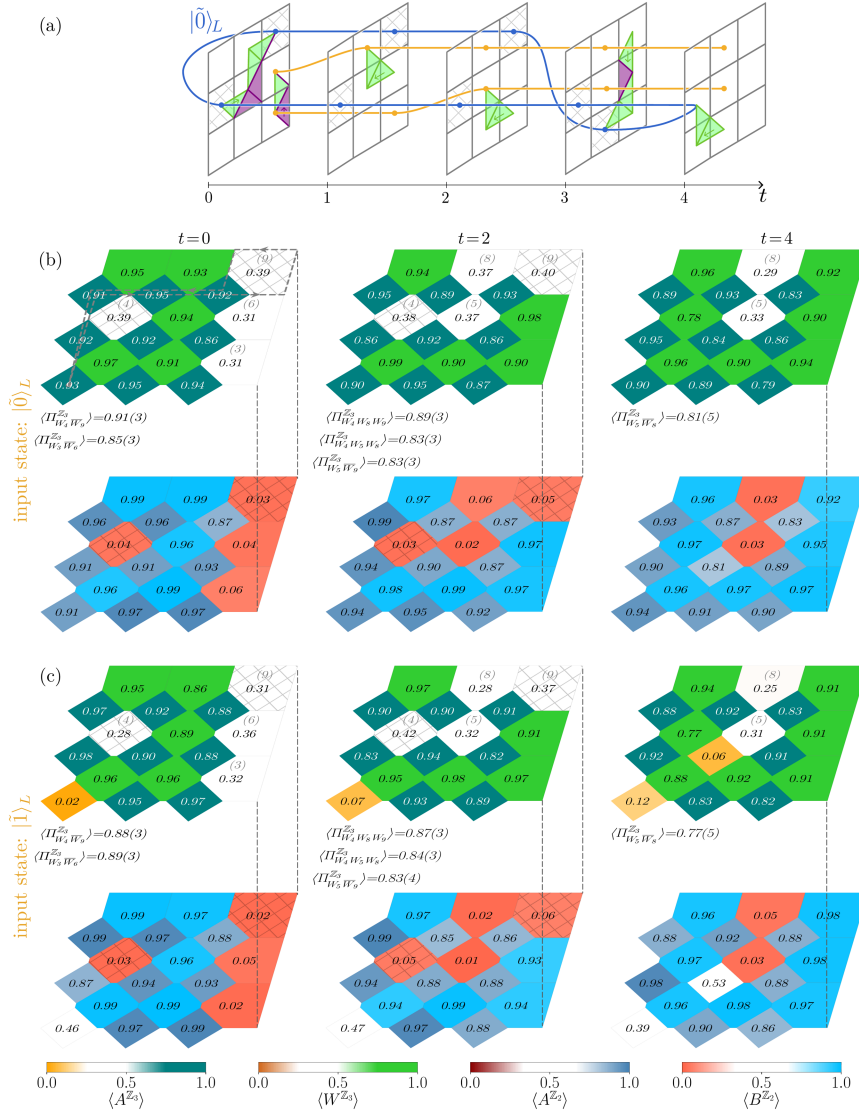


Extended Data Fig. 2 | Ground state preparation using mid-circuit measurements and feed-forward. Expectation values of local and Z-logical projectors are shown. The $\mathbb{Z}_3$ toric-code ground state is prepared unitarily by enforcing plaquette stabilizers (in contrast to the vertex stabilizers used in the main-text experiments), yielding a +1 eigenstate of the $\mathbb{Z}_3$ X-logical operators. This is verified by $Z_{L,\text{hor}}^{\mathbb{Z}_3}$ and $Z_{L,\text{vert}}^{\mathbb{Z}_3}$ values near 1/3. The $\mathbb{Z}_2$ charge-conjugation symmetry is then gauged using an adaptive circuit (Supplementary Information Section S6B1) with mid-circuit measurements and conditional error correction based on stabilizer outcomes. Given that Π is the projector onto the ground-state manifold, $f = \text{Tr}(\rho\Pi)^{1/18}$ denotes the per-qudit fidelity of the prepared state ρ . The average (maximum) standard error for local projectors is 0.012 (0.017), and for Z-logical projectors the average and maximum standard errors are 0.019 and 0.028, respectively.



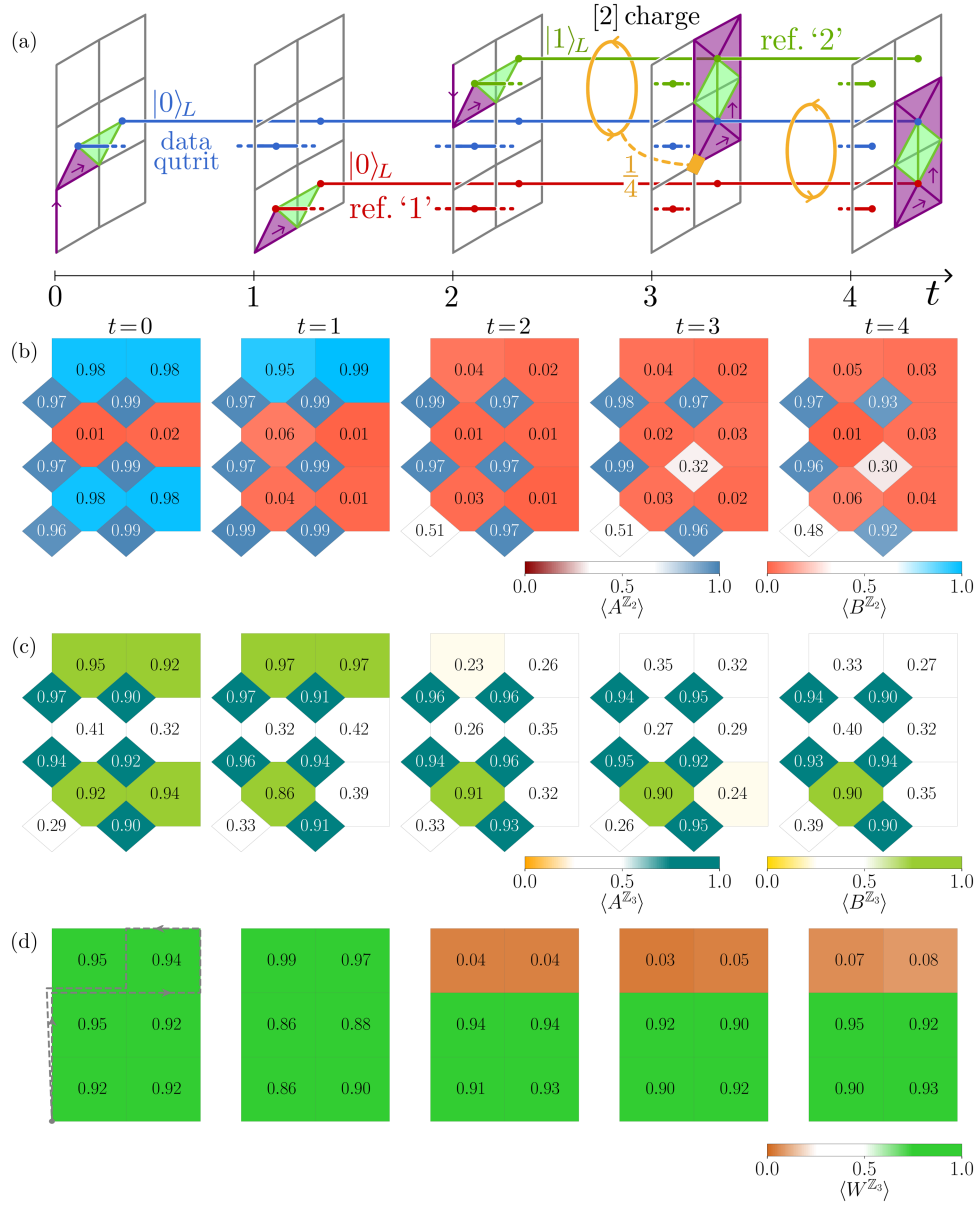
Extended Data Fig. 3 | Pull-through gate protocol. (a) A control qutrit, initialized in the $|0\rangle_L$ state, is shown with blue worldlines; plaquettes hatched by $\times$ indicate the locations of its endpoints at each time slice. The control qutrit is moved around a target qutrit initialized in the $|0\rangle_L$ state and shown with red worldlines. The ribbon operator implementing the motion is constructed from elementary triangle operators (Supplementary Information Section S1D). (b) Measured expectation values of the local projectors $A^{\mathbb{Z}_2}$, $A^{\mathbb{Z}_3}$, $B^{\mathbb{Z}_2}$ and $B^{\mathbb{Z}_3}$,

together with the nonlocal projector $W^{\mathbb{Z}_3}$, shown at different stages of the protocol. For all local and $W^{\mathbb{Z}_3}$ measurements, the average standard error is 0.023, with a maximum of 0.052. Also shown are the W -flux correlators $\Pi_{W_{p_1}W_{p_2}}^{\mathbb{Z}_3}$, which certify inter- and intra-pair delocalized $\mathbb{Z}$ correlations of the control and target qutrits throughout the protocol. In the noise-free limit these correlators equal 1; the measured values are consistent with this expectation, indicating high-fidelity state preparation.



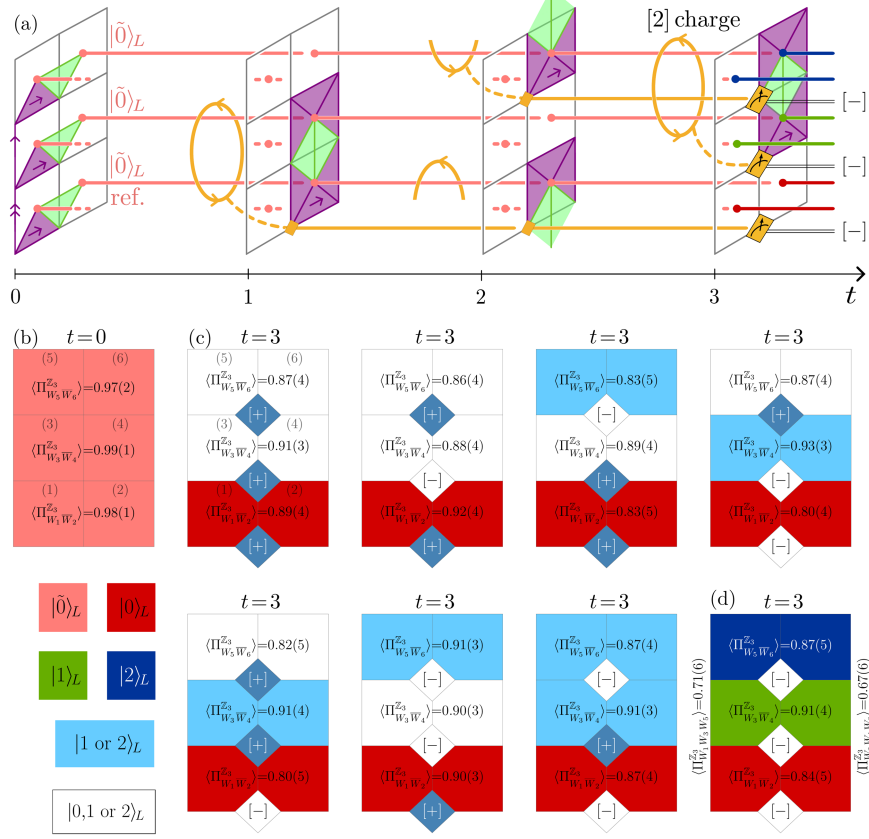
Extended Data Fig. 4 | $\mathcal{X}$ -basis measurement protocol. (a) Spacetime diagram of a ‘measurement’ qutrit braiding around the flux endpoint of an input qutrit. Worldlines are shown in blue (measurement) and yellow (input). Plaquettes marked by $\times$ denote the measurement qutrit flux endpoints on the 3×3 lattice. (b,c) Measurement results for the $\mathbb{Z}_2$ and $\mathbb{Z}_3$ projectors. The input qutrit is prepared in the $|\tilde{0}\rangle_L$ state in panel (b) and in the $|\tilde{1}\rangle_L$ state in panel (c). In each panel, the lower grid shows the local $A^{\mathbb{Z}_2}$ and $B^{\mathbb{Z}_2}$ projectors, while the upper grid shows the local $A^{\mathbb{Z}_3}$ and nonlocal $\Pi_W^{\mathbb{Z}_3}$ projectors (see Supplementary Information Section S2C for precise W -flux paths). The vertex projectors $A^{\mathbb{Z}_2}$

and $A^{\mathbb{Z}_3}$ remain close to +1 at all sites except the origin and the remnant charge site, while the local $B^{\mathbb{Z}_2}$ plaquette stabilizers remain near 0 at qutrit endpoints and reveal no logical information. The nonlocal W -flux projectors, which distinguish the absolute logical states $|0\rangle_L, |1\rangle_L$, and $|2\rangle_L$, remain near $1/3$ due to the superposition of these states. Two-body W -flux correlators $\Pi_{W_{p_1}W_{p_2}}^{\mathbb{Z}_3}$ are close to 1, indicating that the prepared state closely approximates the noiseless ideal. The local $\langle A^{\mathbb{Z}_2}, A^{\mathbb{Z}_3}, B^{\mathbb{Z}_2} \rangle$ and nonlocal $\langle W^{\mathbb{Z}_3} \rangle$ projectors have an average (maximum) standard error of 0.028 (0.072).



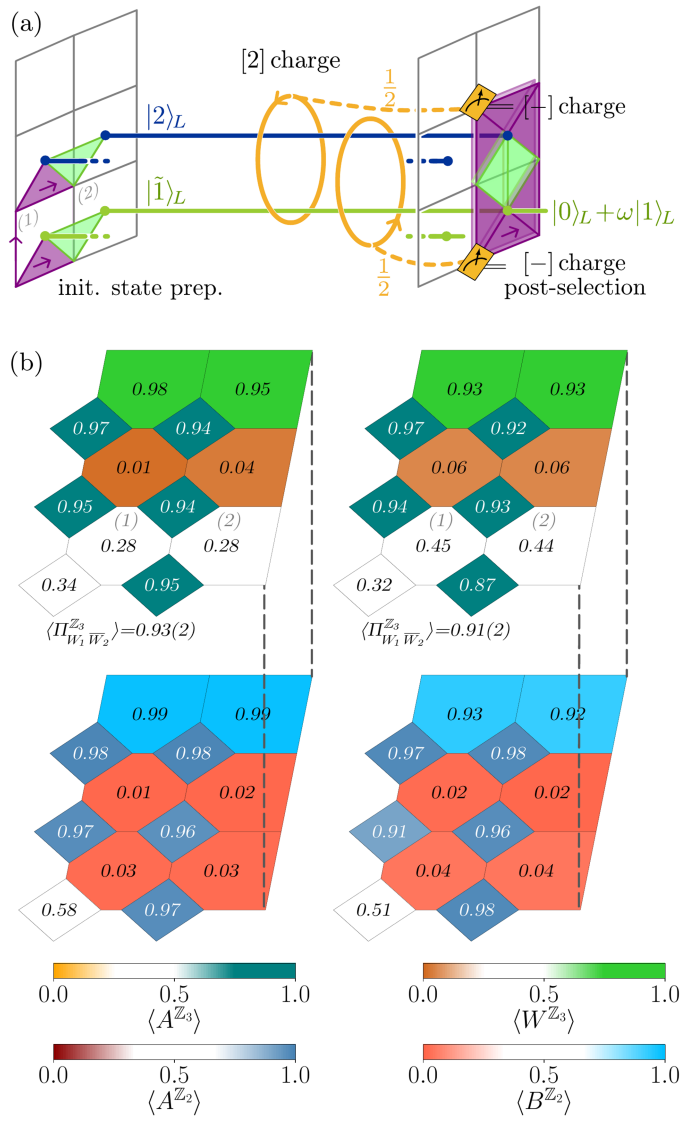
Extended Data Fig. 5 | Z-basis measurement protocol. (a) A data qutrit (blue) and two reference qutrits, labeled '1' (red) and '2' (green), are initialized in $|0\rangle_L$, $|0\rangle_L$, and $|1\rangle_L$ states at $t=0$, $t=1$, and $t=2$, respectively. [2] charge pairs loop around the endpoints of data-reference qutrit pairs ($t=2$ - $t=3$ for data and reference '2' qutrits, $t=3$ - $t=4$ for data and reference '1' qutrits). (b) Expectation values of local $\mathbb{Z}_2$ vertex (A^{Z_2}) and plaquette (B^{Z_2}) projectors. $B^{Z_2} \rightarrow 0$ upon logical qutrits initialization indicate presence of logical qutrit endpoints. The vertex projector at the first [2] charge pair fusion site yields a value $\approx 1/4$, confirming remnant $[-]$ charge. Subsequent [2] charge pair fusion between $t=3$

and $t=4$ leaves A^{Z_2} values unchanged, indicating no additional charge accumulation. (c) $\mathbb{Z}_3$ vertex (A^{Z_3}) and plaquette (B^{Z_3}) projector expectation values. A^{Z_3} remains constant (≈ 1 , except $\approx 1/3$ at origin). B^{Z_3} projectors are measured $\approx 1/3$ at all qutrit initialization sites provide no information about the internal state of the qutrits. (d) Nonlocal W^{Z_3} projector values distinguish internal states: W^{Z_3} for $|0\rangle_L$ state of data and reference '1' qutrits and ≈ 0 for $|1\rangle_L$ state of reference '2' qutrit. Across all local and nonlocal projectors, the average standard error is 0.030, with a maximum standard error of 0.064.



Extended Data Fig. 6 | Z bureau of standards protocol. (a) Schematic of the protocol used to define a logical qutrit in the Z -basis via relative flux comparisons. Three C_2 flux pairs are initialized in the $|\tilde{0}\rangle_L$ state, with the lowest pair chosen as the reference $|0\rangle_L$. The logical states of the remaining pairs are determined through three comparison measurements implemented by braiding a [2] charge pair. (b–d) Experimental verification of the prepared states. (b) At $t=1$, the $\Pi_{W_{p1}W_{p2}}^{Z_3}$ projectors take values close to unity, demonstrating that the prepared states closely approximate the ideally prepared (i.e., noiseless) $|\tilde{0}\rangle_L$ pairs. (c) The data is shown for different sets of

'partially' prepared basis states at $t=3$, where each set is defined by post-selection on a specific outcome of the remnant charge (A^{Z_2}) measurement. (d) The final panel at $t=3$ displays three distinct states where all three remnant charges are measured in $[-]$ states, post-selected with an acceptance rate of ~ 0.115 , comparable with the theoretical prediction of 0.125 ; the total acceptance rate (including heralding) is ~ 0.06 . Strong intra-pair and inter-pair Z correlations are confirmed by measurements of $\Pi_{W_{p1}W_{p2}}^{Z_3}$ and $\Pi_{W_{p1}W_{p2}W_{p3}}^{Z_3}$, respectively.



Extended Data Fig. 7 | Magic state preparation protocol. (a) Schematic illustrating the creation of a magic state by inducing projection via measurement (Supplementary Information Section S10E). (b) Expectation values of local projectors A^{z_2} , A^{z_3} , and B^{z_2} , and the nonlocal projector W^{z_3} after initial state preparation and magic state preparation. For plaquettes (1) and (2), the measured $\Pi_{W_1 W_2}^{z_3}$ values are 0.28 and 0.28 after initial state preparation, in close agreement with the theoretical value of $1/3$. Following the complete protocol, these values change to 0.45 and 0.44, consistent with the theoretical value of $1/2$. The high fidelities of the prepared states is further supported by measurements of $\Pi_{W_1 W_2}^{z_3}$, which are close to unity. Across all W^{z_3} and local projector measurements, the average standard error is 0.017, and the maximum standard error is 0.057.

Extended Data Table 1 | Circuit specifications

Experiment	Lattice size	No. of native one-qubit gates	No. of native two-qubit gates	Depth (two-qubit gates)
Ground state unitary prep.	3×3	429	238	43
Ground state measurement-based prep.	3×3	545	274	50
Single anyon state prep.	3×3	453	308	76
$\mathcal{Z}$ -basis measurement at $t = 4$	3×2	423	269	74
$\mathcal{Z}$ -basis bureau of standards at $t = 3$	3×2	646	437	107
Magic state preparation	3×2	507	346	134
$\mathcal{X}$ -basis measurement at $t = 5$	3×3	938	744	232
Pull-through gate at $t = 6$	3×3	1,008	845	307

Gate counts and depths for the circuits used in each experiment. Destructive measurement at the end of the circuit was performed to compute the plaquette projectors. Furthermore, each circuit was optimized using local qutrit basis transformations (Supplementary Information Section S5). For reference, the execution time for a single shot of the $\mathcal{X}$ -basis measurement at $t = 5$ was approximately 5.0 s, and for the pull-through gate experiment at $t = 6$ it was approximately 5.9 s on Quantinuum’s H2-1 quantum computer.